



Project FORESIGHT

AI-Powered Demand & Inventory Intelligence Platform

Project Report

Submitted by

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Retail sales, demand forecasting and inventory analysis
A twelve-page Power BI dashboard project

1 Project Overview

Project FORESIGHT brings retail sales, demand patterns and inventory analysis into one Power BI reporting project. Its twelve dashboard pages help a reviewer understand business performance, identify stock concerns and prioritise follow-up actions. The report moves from an executive summary to product, category, promotion, seasonality and customer analysis.

The project name and subtitle follow the supplied example. The submitted dashboard demonstrates a business intelligence and forecasting interface; the underlying AI or machine-learning implementation has not been verified from the PDF export.

Objectives

- Understand revenue, sales volume and product profitability.
- Explore demand across seasons, promotions and customer segments.
- Identify possible stockout, overstock and slow-moving inventory issues.
- Present forecast outputs and assess their suitability for planning.
- Bring business findings together in an executive recommendation page.

2 Problem Statement

A retail business can carry too much of one product while running short of another. Excess stock ties up working capital, while shortages can lead to missed sales. Sales and inventory records need to be viewed together so that demand patterns can inform purchasing, stock review and clearance decisions.

3 Data Description

The main sales reporting period is 1 January 2024 to 31 December 2025. The forecast page extends to 31 March 2026. Values below are taken from the submitted dashboard export.

Item	Detail
Revenue and units	3.10 billion revenue and 511,810 units sold
Gross profit	Approximately 1.22 billion displayed
Categories	Home Decor, Furniture, Storage, Kitchen and Lighting
Customers	Approximately 2K displayed; High Value, Regular and Occasional segments
Inventory fields	Stock, reorder point, lead time, inventory value and risk status
Calendar fields	Date, month, quarter, season, weekday and holiday indicators

The product count and full SKU coverage require confirmation in the source files. Monetary amounts are stated without a currency because formatting differs across the dashboard pages.

4 Technology Stack

Layer	Tools	Purpose and status
Source data	Excel	Project setup uses sales, product, inventory and calendar workbooks.
Preparation	Power Query	Intended import and cleaning layer; query steps are not visible in the PDF.
Analytics	Power BI Desktop and DAX	Dashboard creation and business measures. Power BI Desktop is visible in the export.
Forecasting	Method to be confirmed	Forecast charts and bounds are visible; model code and validation results were not supplied.
Documentation	Word and PDF	Editable project report and a fixed-layout submission copy.

The example also lists Python, SQL, GitHub and deployment tools. These are suitable potential extensions, but they are not recorded here as completed components without supporting project files or links.

5 System Architecture

The intended architecture starts with business tables, prepares them for analysis, connects them in a Power BI model and presents the results through dashboard pages. The export shows the reporting layer; the connections below describe the logical design to verify in the editable project.

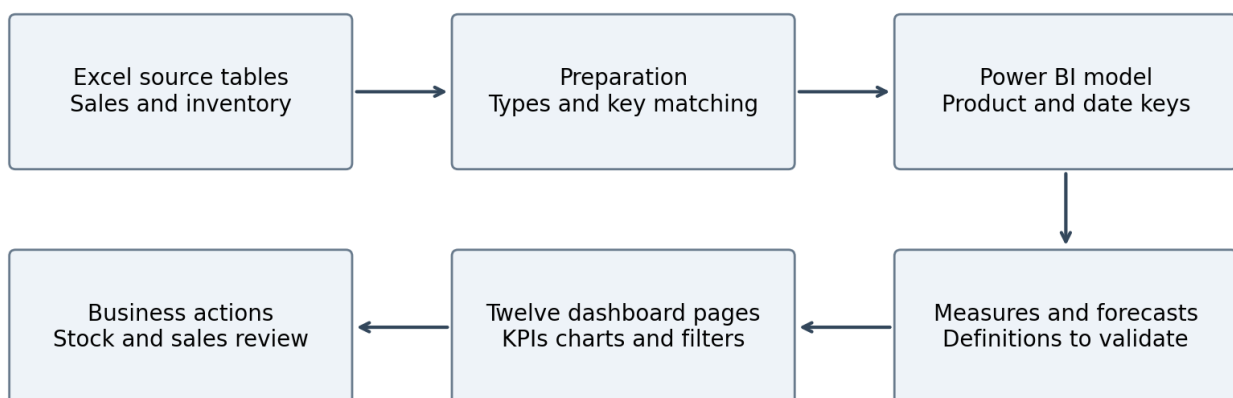


Figure 1 Conceptual system architecture

Model relationships

A unique SKU in the product master should filter related sales and inventory records. A unique calendar date should filter sales and inventory snapshot dates. Sales records and stock snapshots should retain their own level of detail. Customer identifiers support segmentation, although the PDF does not reveal where customer attributes are stored.

The same measure definitions and date filters should be used across pages so that executive totals agree with detailed analysis.

6 Methodology

6 1 Data preparation

The project setup uses sales_daily.xlsx, sku_master.xlsx, inventory_snapshots.xlsx and calendar.xlsx. The intended preparation process checks data types, duplicates, missing values and matching SKU keys. Product and category mapping is particularly important because blank category groups remain visible in the inventory pages.

6 2 Demand forecasting

The Forecast Dashboard presents actual revenue, a forecast line, bounds labelled 95%, category forecasts and a 30-day outlook. The specific forecasting algorithm and holdout validation are not documented in the supplied files. Actual and forecast values must be aligned to the same dates, units and aggregation level before accuracy is assessed.

The displayed MAPE value of 97.94 and variance of 4.27M need review. They should not be presented as evidence of a validated forecasting model until their formulas and formats are confirmed.

6 3 Stockout risk

The dashboard groups stock records into CRITICAL, WARNING and SAFE. A useful risk rule compares current stock and expected demand with the time needed to replenish. The precise thresholds and whether the counts represent unique SKUs or repeated snapshots must be confirmed in the Power BI model.

6 4 Overstock and slow-moving stock

The overstock page presents excess value, months to clear, annual holding cost and a discount scenario. These indicators are intended to identify stock that could be reduced through purchasing changes or targeted clearance. Their calculations need a documented stock target, valuation basis and sales-rate period.

6 5 Recommendations

The final dashboard combines growth, profitability, stock indicators, a risk matrix and an opportunity assessment. It lists actions such as reordering critical products, reducing excess inventory and reviewing weak products. Operational priorities should use validated measures and a consistent reporting period.

6 6 Quality checks

- Reconcile revenue and units with the source workbooks.
- Check unmatched SKU keys and use a defined inventory snapshot date.
- Verify ratio measures under identical filters across dashboard pages.
- Test forecast accuracy on a period excluded from model fitting.
- Check chart titles, gauge limits and percentage formatting before export.

This report reviews the static PDF. It does not verify live slicer behaviour, refresh operations or the exact DAX expressions.

7 Dashboard Walkthrough and Insights

7. 1 Executive Dashboard

Cards summarise revenue, gross profit and units. The daily trend, category share and product table provide context.

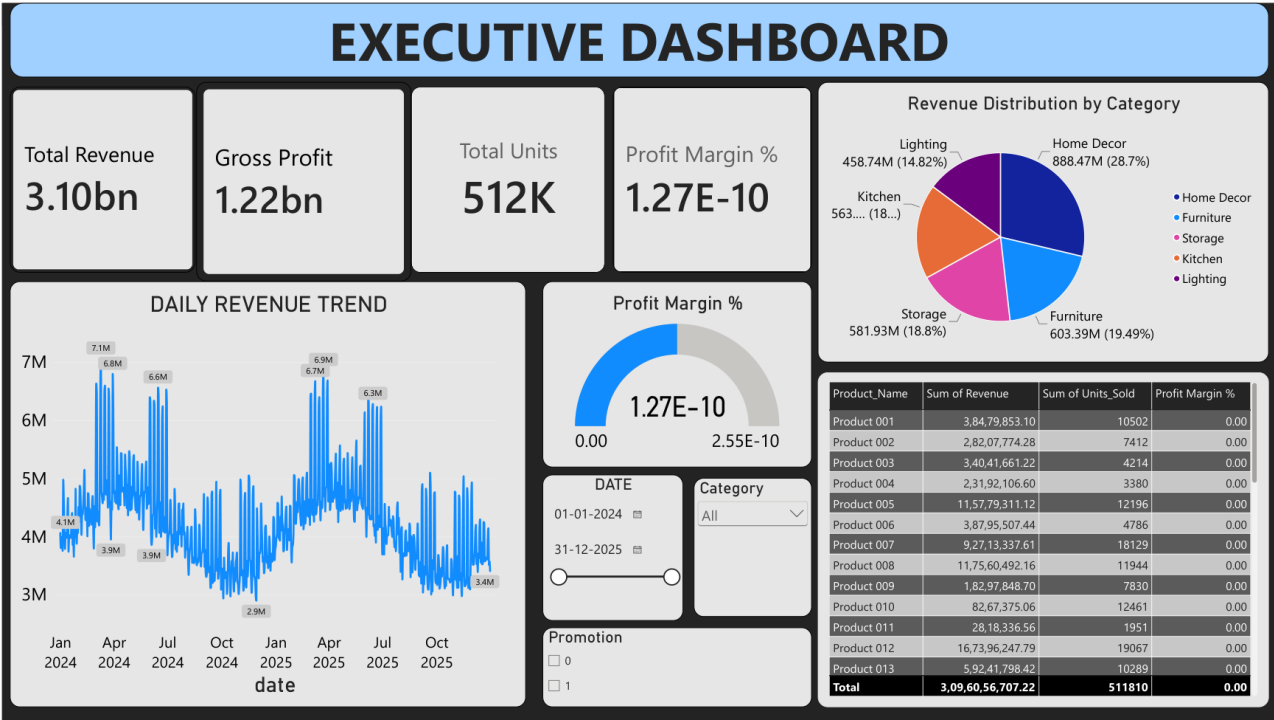


Figure 2 Executive Dashboard | Source dashboard page 1

Main observations

- The executive table shows revenue of 3,096,056,707.22 and 511,810 units sold. Gross profit is displayed as approximately 1.22 billion.
- Home Decor contributes 28.70% of revenue, followed by Furniture, Storage, Kitchen and Lighting.

Interpretation and review

The near-zero profit margin display needs reconciliation with the profitability shown elsewhere.

7. 2 Sales Analytics Dashboard

The page compares revenue over time, weekdays and promotion status, with a monthly matrix for annual comparisons.

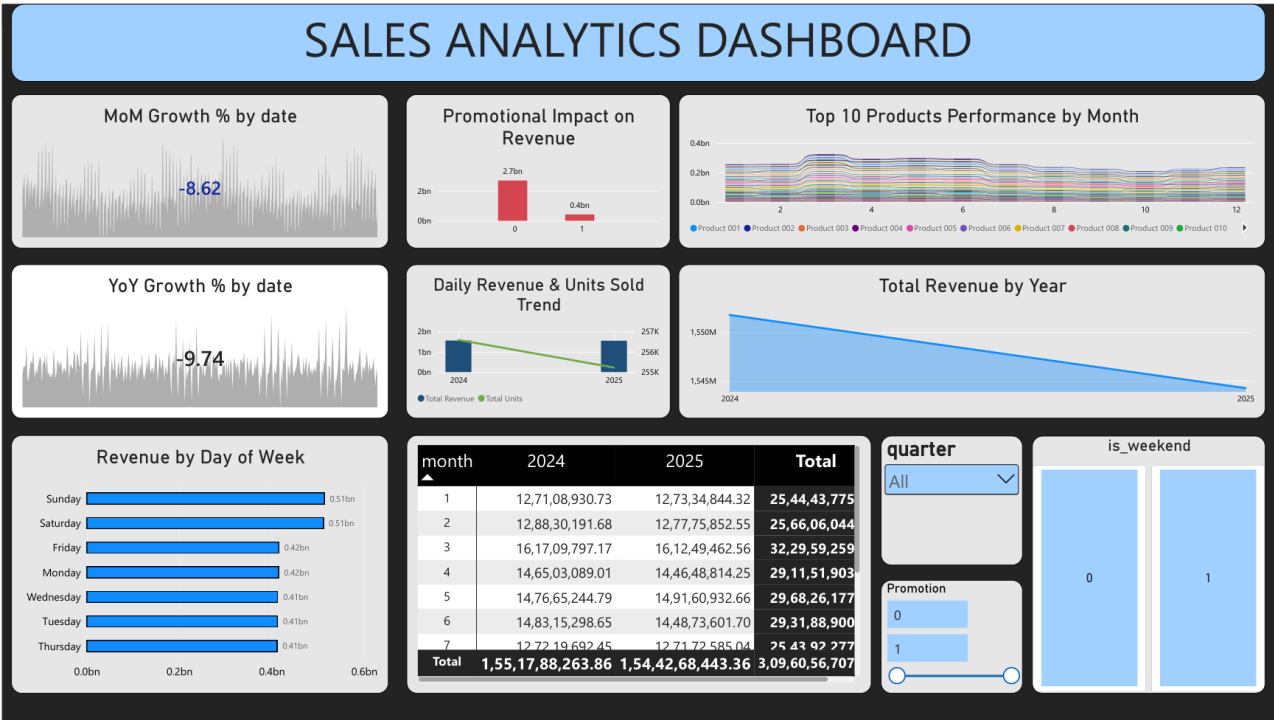


Figure 3 Sales Analytics Dashboard | Source dashboard page 2

Main observations

- The annual matrix reports 1,551,788,263.86 for 2024 and 1,544,268,443.36 for 2025, implying an annual decline of about 0.48%.
- Saturday and Sunday each contribute approximately 0.51 billion in the weekday chart.

Interpretation and review

The year totals support the approximately 0.48% decline calculated in this report; growth cards need a consistent period definition.

7.3 Product Performance Dashboard

Revenue ranking, margin comparison and a volume-versus-profitability chart support product review.

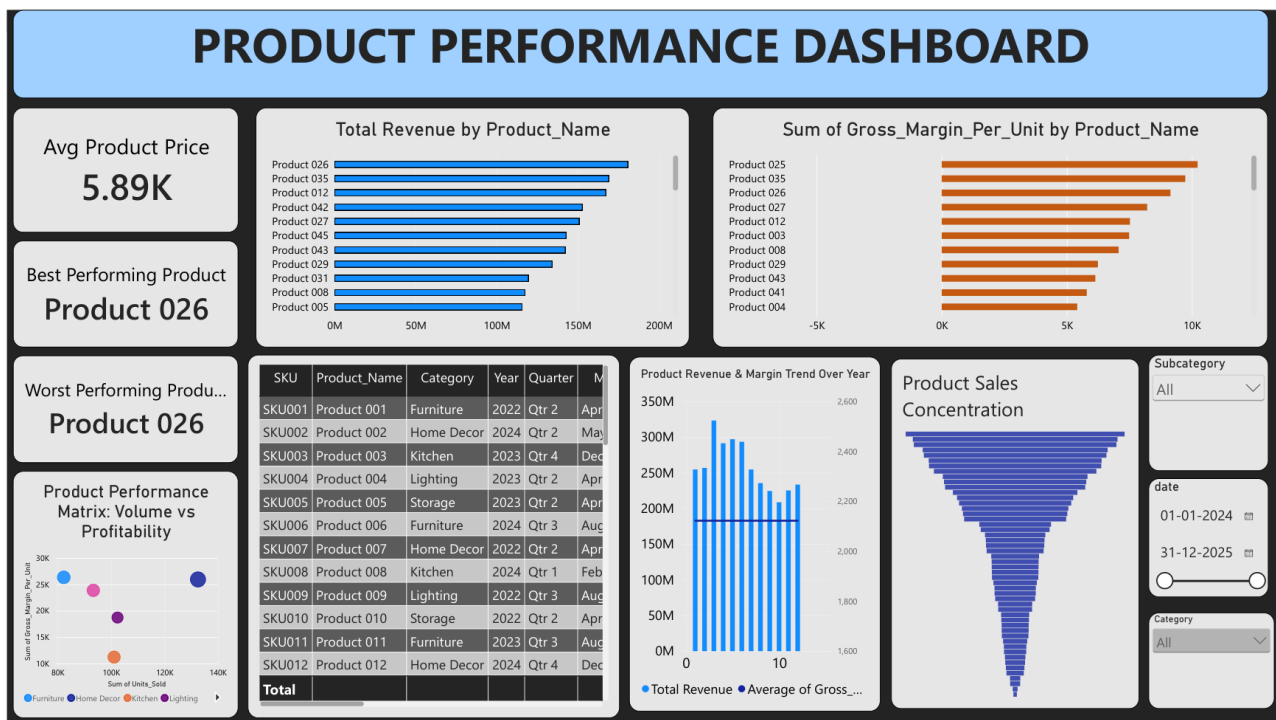


Figure 4 Product Performance Dashboard | Source dashboard page 3

Main observations

- Product 026 leads the displayed revenue ranking, followed by Product 035 and Product 012.
- The margin and volume visuals help compare sales scale with profitability rather than ranking products only by revenue.

Interpretation and review

The best and worst product cards both show Product 026 and should be checked.

7 .4 Category Performance Dashboard

Category revenue, contribution and profitability are presented with season, quarter and holiday filters.

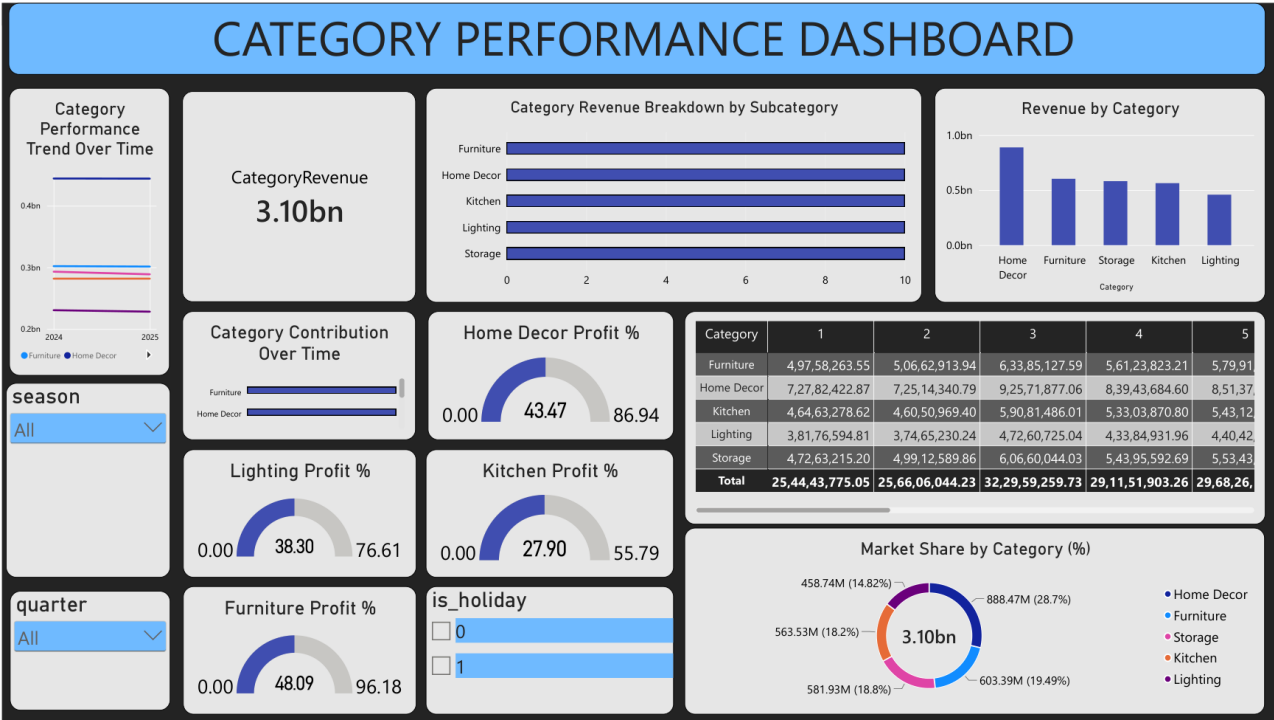


Figure 5 Category Performance Dashboard | Source dashboard page 4

Main observations

- Category revenue is approximately 888.47M for Home Decor, 603.39M for Furniture, 581.93M for Storage, 563.53M for Kitchen and 458.74M for Lighting.
- The page includes category margin gauges and a monthly revenue matrix for more detailed comparisons.

Interpretation and review

The chart labelled market share represents share of the project revenue unless external market data is supplied.

7.5 Inventory Dashboard

Stock units, value, age and turnover are shown with category and product breakdowns.

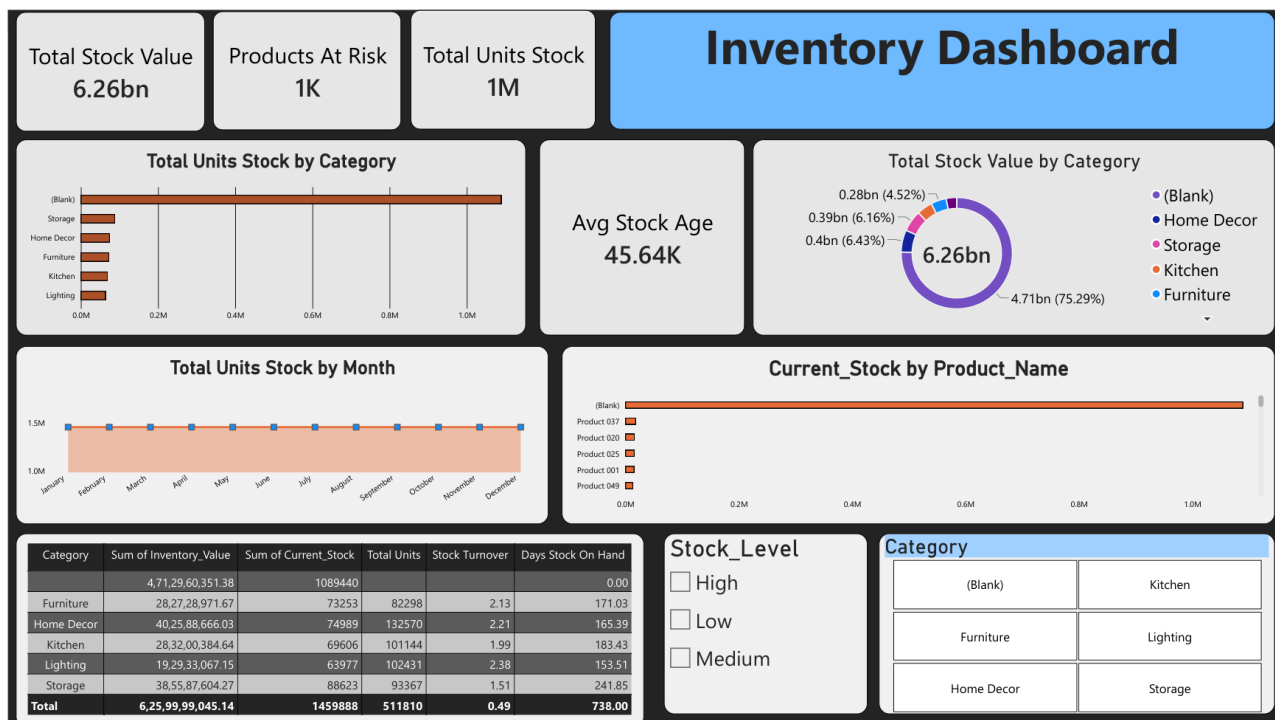


Figure 6 Inventory Dashboard | Source dashboard page 5

Main observations

- The displayed inventory total is approximately 6.26 billion in value and 1,459,888 stock units. These totals require snapshot validation.
- Blank carries about 75.29% of inventory value, which materially limits category interpretation.

Interpretation and review

Blank accounts for a large share of stock value. Resolve product mapping and snapshot aggregation before using the totals.

7.6 Stockout Risk Dashboard

Risk cards and detailed stock tables compare critical, warning and safe classifications.

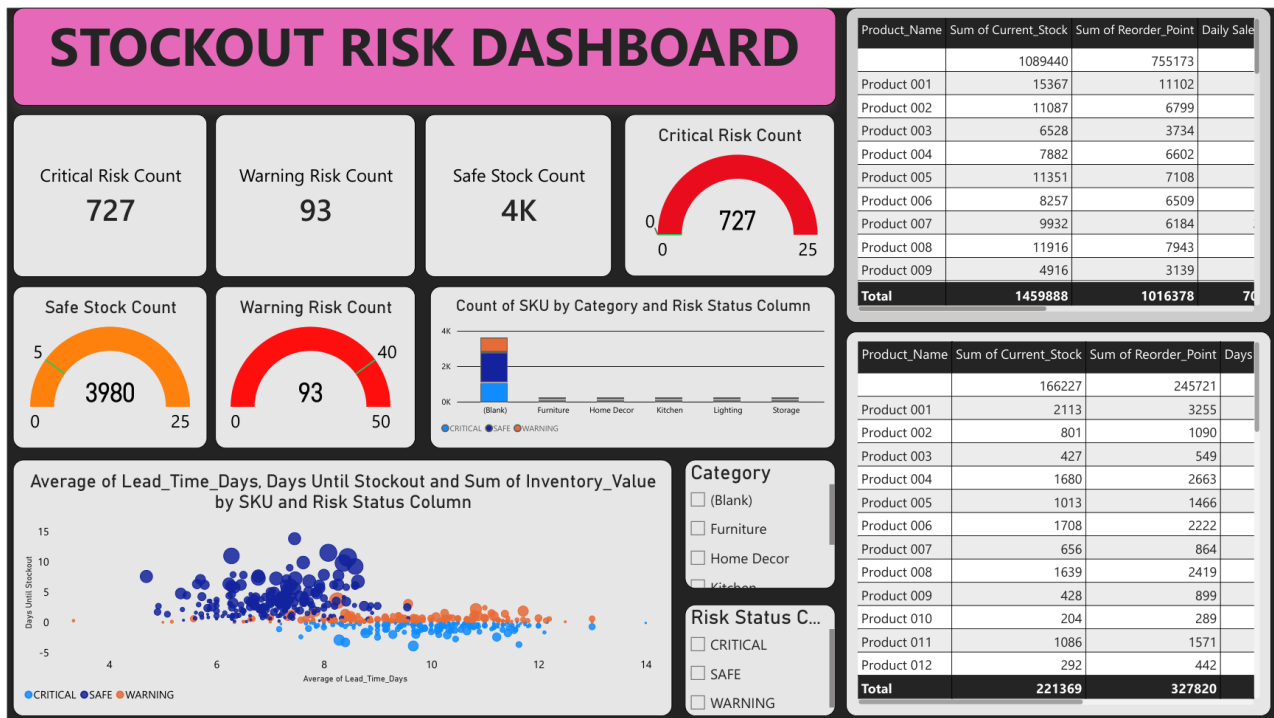


Figure 7 Stockout Risk Dashboard | Source dashboard page 6

Main observations

- The visible risk counts are 727 CRITICAL, 93 WARNING and 3,980 SAFE records. These should not be described as distinct products without checking the measure.
- Stock and reorder-point tables support detailed replenishment review.

Interpretation and review

Risk counts need a documented counting grain and gauge scales that contain the displayed values.

7.7 Overstock Dashboard

The page brings together excess value, holding cost, slow-moving products and clearance comparisons.

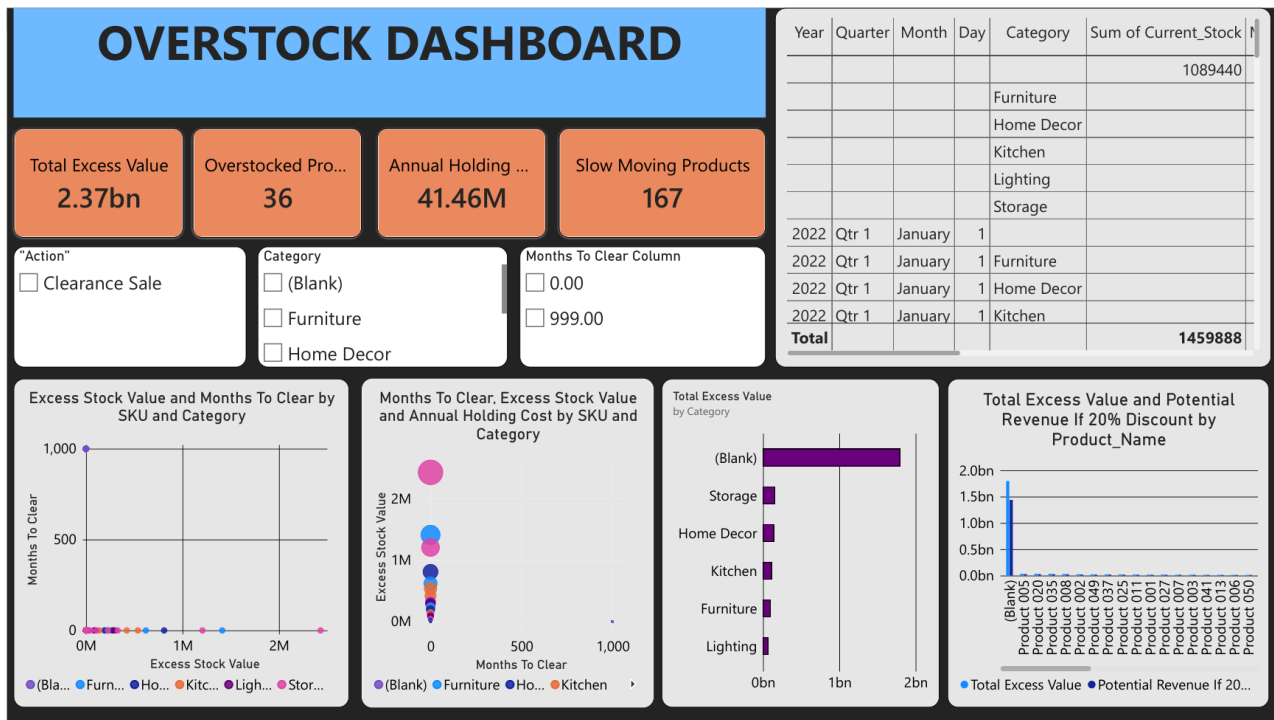


Figure 8 Overstock Dashboard | Source dashboard page 7

Main observations

- Cards show 2.37 billion excess value, 36 overstocked products, 167 slow-moving products and 41.46M annual holding cost.
- The differing counts and months-to-clear values need consistent definitions before estimating a clearance plan.

Interpretation and review

Confirm the excess-stock target and treatment of the 999 months-to-clear value before using clearance estimates.

7.8 Promotion Dashboard

Monthly and category comparisons separate promotional and non-promotional sales. Product detail shows revenue and profit.

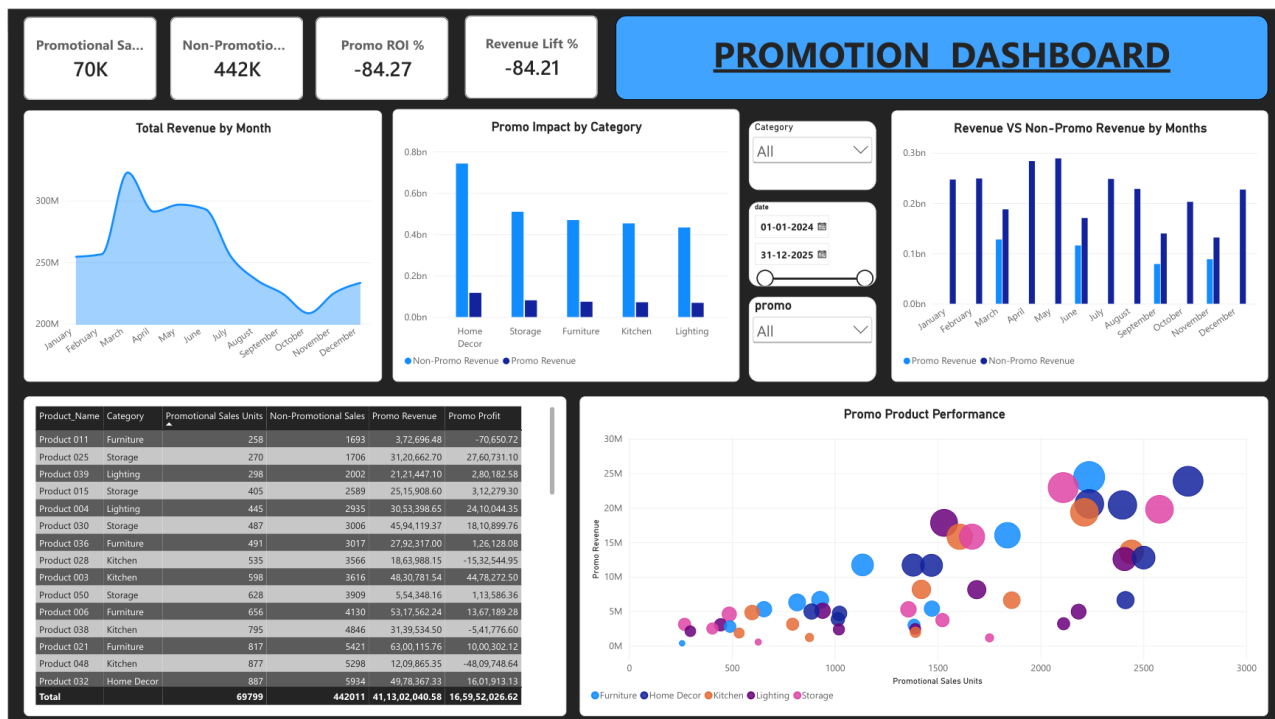


Figure 9 Promotion Dashboard | Source dashboard page 8

Main observations

- Promotional units total 69,799, compared with 442,011 non-promotional units.
- Promotional revenue is 411,302,040.58, or about 13.28% of the overall displayed revenue. Unequal exposure limits conclusions about campaign effectiveness.

Interpretation and review

Revenue differences reflect unequal sales volumes; the displayed lift and ROI are not proof of causal promotion impact.

7.9 Seasonality Dashboard

Seasonal and monthly charts show how demand varies across the calendar, with a holiday detail table.

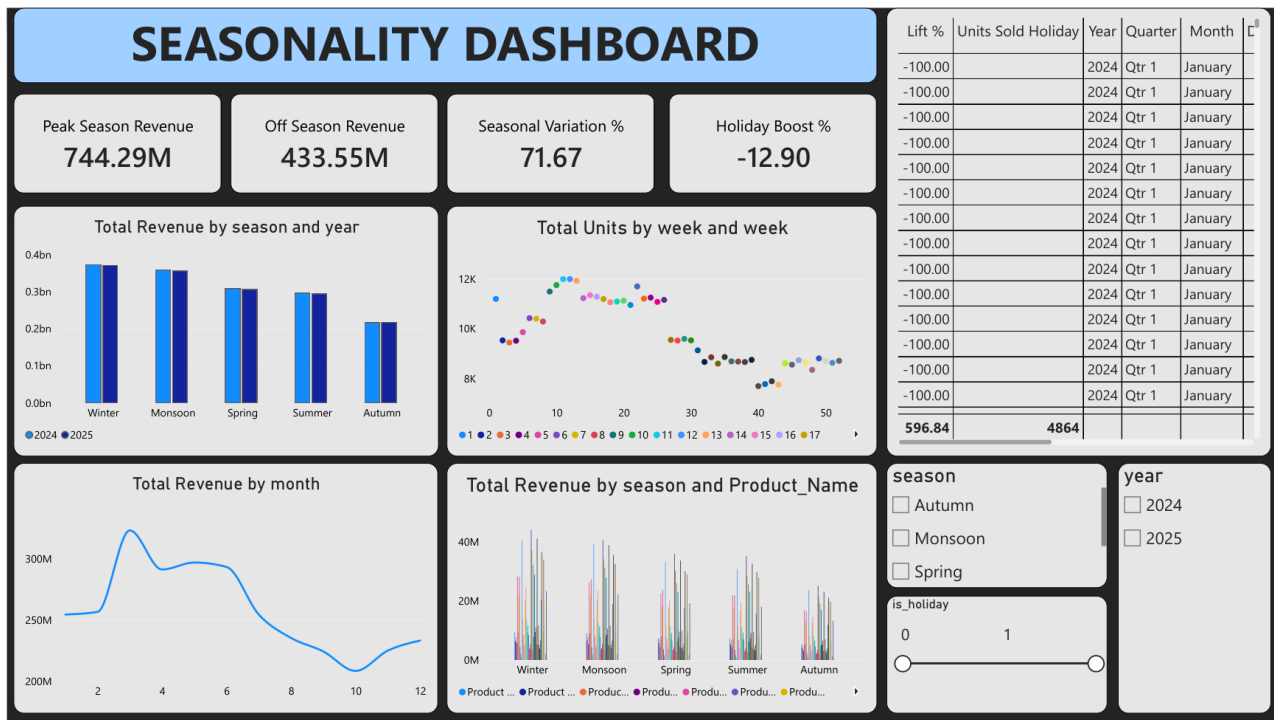


Figure 10 Seasonality Dashboard | Source dashboard page 9

Main observations

- Cards show peak-season revenue of 744.29M and off-season revenue of 433.55M, with a displayed variation of 71.67%.
- Monthly, weekly and seasonal views provide different ways to inspect recurring demand patterns.

Interpretation and review

Season lengths, holiday baselines and missing table values need checking before interpreting seasonal uplift.

7.10 Forecast Dashboard

Actual revenue, forecast lines, confidence bounds and error indicators are displayed together.

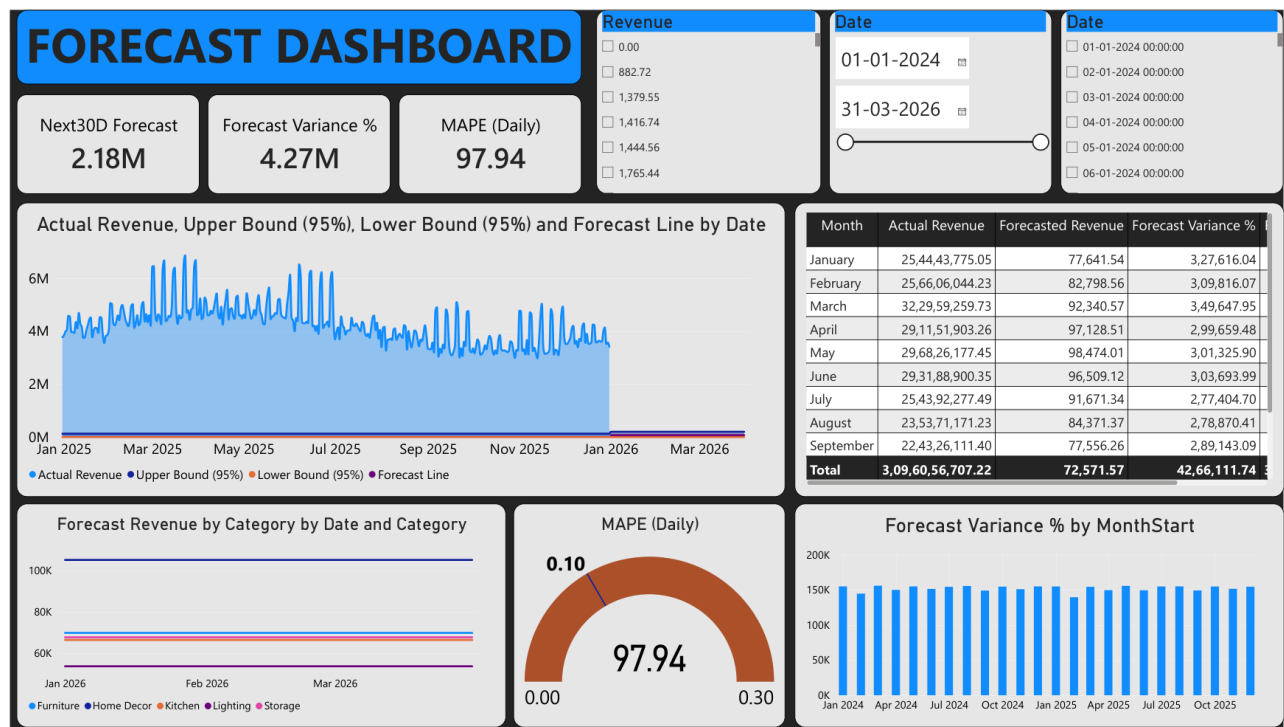


Figure 11 Forecast Dashboard | Source dashboard page 10

Main observations

- The page shows a 2.18M next-30-day forecast and includes actual-versus-forecast and category forecast views.
- Forecast bounds are labelled 95%, but the statistical method and interval coverage are not documented in the export.

Interpretation and review

The forecast output should be treated as experimental until time grain, scaling and holdout accuracy are validated.

7.11 Customer and Business Insights

Customer segments, purchase frequency, a Pareto view and a category matrix describe customer contribution.



Figure 12 Customer and Business Insights | Source dashboard page 11

Main observations

- The segment chart shows 40% Occasional, 40% Regular and 20% High Value. The High Value customer card displays 490.
- Regular customers contribute the largest total segment revenue in the chart. Purchase frequency and revenue per customer add context.

Interpretation and review

Validate the customer count, segment rules and the 100% retention calculation against the intended customer cohort.

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7.12 Executive Recommendation Dashboard

Business health cards, a risk matrix, opportunities and ranked actions summarise the project.

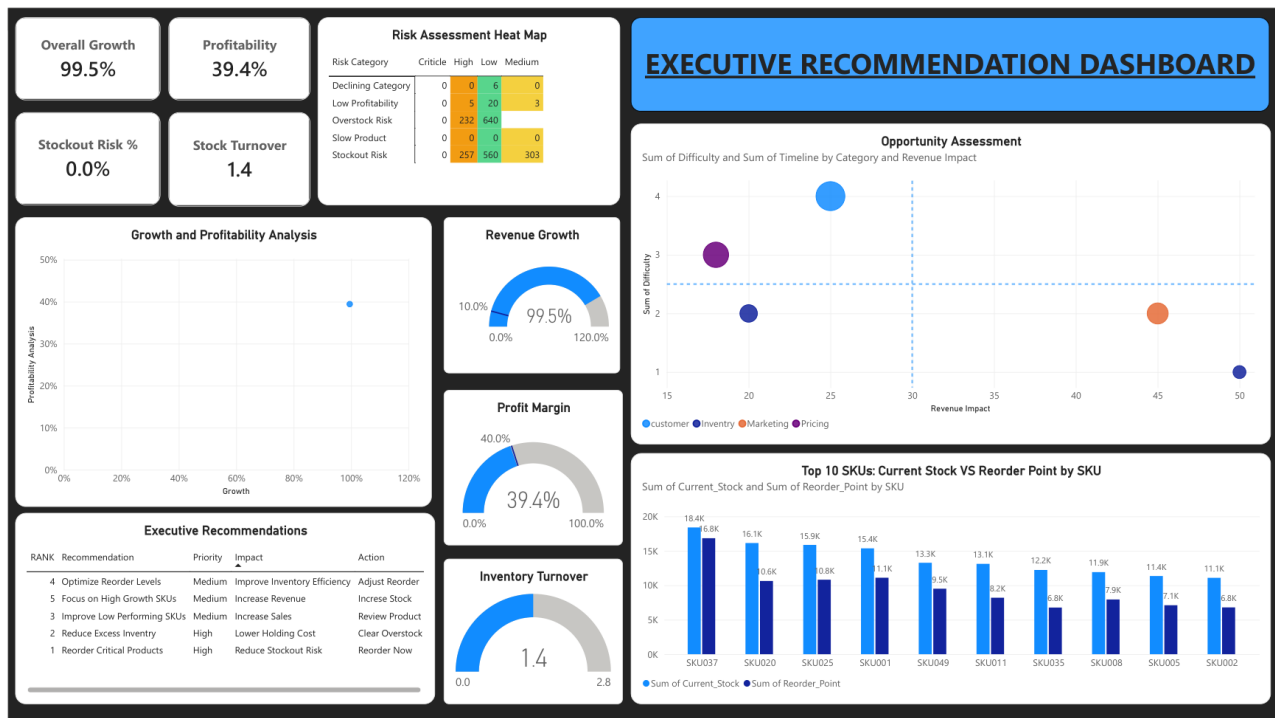


Figure 13 Executive Recommendation Dashboard | Source dashboard page 12

Main observations

- The page displays 39.4% profitability, 1.4 stock turnover, 99.5% growth and 0.0% stockout risk. Some of these values conflict with other pages.
- The recommendation table prioritises critical-product reordering, excess-stock reduction and product performance review.

Interpretation and review

Reconcile growth and stockout measures with the detailed pages before prioritising actions from this view.

8 Key Insights and Business Recommendations

The strongest supported findings are the scale of recorded sales, Home Decor's leading revenue share and the larger volume of non-promotional sales. Stock and forecast measures need validation before they determine purchasing or clearance quantities.

Priority	Recommendation	Expected benefit
High	Resolve SKU mapping and reconcile key measures across pages.	More reliable product, category and risk reporting.
High	Review critical stock records using the latest stock and supplier lead times.	Reduce avoidable shortages once risk rules are confirmed.
High	Validate excess-stock quantities and select appropriate clearance items.	Release working capital and reduce holding costs.
Medium	Review low-revenue products alongside their margin and demand.	Improve product mix and purchasing decisions.
Medium	Adjust stock targets to validated seasonal demand patterns.	Match replenishment more closely to demand.
Medium	Compare promotion results over similar exposure and include campaign costs.	Improve the basis for evaluating promotional spending.

9 Challenges Faced

The following issues are visible in the export or arise directly from documenting it. They identify the main areas for further project refinement.

- Product mapping: a large Blank inventory category limits the stock analysis.
- Measure consistency: profit margin is near zero on the executive page but 39.4% on the recommendation page.
- Period comparison: the 99.5% growth card differs from the roughly 0.48% decline calculated from annual totals.
- Risk aggregation: critical counts and the 0.0% risk card need a shared definition and counting level.
- Forecast comparability: actual and forecast values, error units and gauge scales need reconciliation.
- Presentation: several labels are truncated, and some risk values exceed gauge limits.

10 Conclusion and Future Scope

Project FORESIGHT presents a structured approach to retail sales and inventory analysis through twelve Power BI pages. The report connects overall performance with product contribution, category demand, inventory conditions, promotions and customer behaviour. Its final recommendation page provides a practical place to organise management actions.

The current dashboard establishes a useful reporting framework. The next step is to strengthen the underlying measures and product mapping so that the executive indicators agree with the detailed pages. Forecasting and stock recommendations will become more dependable when their assumptions and validation results are documented.

Future scope

- Complete SKU coverage and maintain one consistent product master.
- Develop and validate SKU-level demand forecasts against a simple baseline.
- Add suggested reorder quantities using demand, lead time and safety stock.
- Introduce scheduled refresh and critical-stock alerts after deployment is configured.
- Evaluate pricing and promotions with comparable periods and cost information.
- Maintain a data dictionary and measure catalogue for future users.

11 Project Links

No project-specific repository, deployment or video links were supplied. Add the verified links below before final submission if they are required.

Deliverable	Link or status
Source code repository	Not provided
Live deployment	Not provided
Demo video	Not provided
Feedback video	Not provided

Sources and editable files

Dashboard evidence: final Dashboard(1).pdf, 12 pages, supplied by MD Shoaib Akhtar. Reference structure and project naming: Project_FORESIGHT_Report.pdf.pdf. Reference-only personal details and unverified technology claims have not been assigned to this project.

The Word report contains editable text and tables. Dashboard screenshots are embedded images; edit the underlying charts in Power BI and replace the screenshots after a fresh export.